

12 (long)

(a) In Fig. 7, an aluminium bar of cross-sectional area 1 cm^2 is suspended between the pole pieces of a magnet that produces a uniform magnetic flux density $B = 0.5 \text{ T}$. In order to insert the bar into the gap between the pole pieces a force F is required as shown. It may be assumed that the energy stored in a magnetic field per unit volume is given by $B^2/2\mu$, where, for aluminium, $\mu = 1.000023 \mu_0$. By equating the work done by the force F to the change in stored magnetic energy, calculate the value of F . [10]

(b) A plane circular search coil is used to measure the magnetic flux density B . The coil consists of one turn, of area 1 cm^2 , and the total resistance of the coil is $R = 1 \text{ k}\Omega$. Calculate the magnetic flux threading the coil, if it is placed between the pole pieces with the plane of the coil perpendicular to the magnetic field. Also, calculate how much charge flows through the coil as it is removed from the field to a place where the magnetic flux density is zero. How could you measure the charge? [8]

(c) An alternative method of measuring B would be to have the search coil rotating about an axis through one diameter. Show that the flux cutting the coil is sinusoidal with time, and calculate the emf if the coil rotates at 3600 revolutions per minute when it is placed between the pole pieces. [8]

(d) In practice, which method is more useful for validating the design of a magnet, and why? [4]

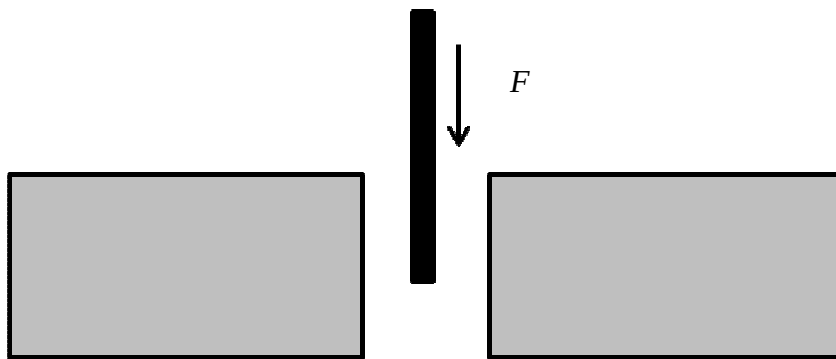


Fig. 7

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Answers

1. (b) $R_1 \times R_4 = \frac{L}{C}$ (c) $\omega = \frac{1}{C\sqrt{R_3 \times R_4}}$
2. (c) Gain $\cong 10$, $f \cong \frac{1}{2\pi CR_3} \cong 21 \text{ MHz}$
3. (a) $40 \text{ k}\Omega$, $2.5 \text{ k}\Omega$ (b) -11.5
4. 0.525 A
5. (a) 400 W , 246 V (b) 1.35 mF
6. (b) $Y = \overline{\overline{A+C+B+C}}$
7. (b) $Y = \overline{\overline{A.D.C.E.B.E}}$
8. (a) $2^{33} \text{ bits} = 2^{30}/10^6 \text{ Mbytes}$
(b) (i) Decrement until zero (ii) Infinite loop
9. (c) $J_A = I.Q_B$, $K_A = \overline{I}.Q_B + I.\overline{Q_B}$, $J_B = \overline{I}$, $K_B = I$
(d) (i) 2 bistables (ii) 2 bistables
10. (c) Vertically downwards
11. (a) $B = \frac{\mu_0 I a^2}{2(z^2 + a^2)^{3/2}}$
12. (a) $2.28 \times 10^{-4} \text{ N}$ (b) $5 \times 10^{-5} \text{ Wb}$, $0.05 \mu\text{C}$ (c) 0.019 V (amplitude)